

5th Grade Team Exam

2026 James Clemens Math Tournament

INSTRUCTIONS:

1. Your team has 60 minutes to complete this examination.
2. One official team from each school consisting of four students and other unofficial teams may compete. Only the scores from the official team will be counted towards a school's final score.
3. This exam consists of 15 free-response questions, each worth 6 points if answered correctly and 0 points if answered incorrectly or left unanswered.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Write your answers to the questions in the provided team answer sheet. You may use the test booklets for scratch work, but only the answers marked in the team answer sheet will be counted. Please write **legibly**.
6. **All answers to the team exam will be a positive integer from 1 to 999, inclusive.**
7. **Leave out** all units and special characters when recording your answer on the team answer sheet. For example, if you get \$1 as your answer, write 1 on the answer sheet.
8. A school's final score for a grade will be calculated by adding the top 4 written test scores in that grade (excluding tiebreaker questions) as well as the team round score of the official team, for a maximum of 330 points.
9. In the event of a tie, the school with the highest team round score will be favored. If team round scores result in another tie, answers will be evaluated starting backwards from question 15 to 1 to determine a winner.
10. Diagrams are not necessarily drawn to scale.
11. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
12. You may keep the test booklet after the exam.

DO NOT TURN THIS PAGE UNTIL TOLD TO DO SO

1. Solve: $11 \cdot 11 + 22 \cdot 11 + 33 \cdot 11$

2. If $(x + 1)^2 - 3 \cdot 5 = 1$, what is the positive value of x ?

3. How many rectangles are present in the shape?



4. A recipe for a dozen cookies requires $1\frac{1}{2}$ cups of flour, 1 egg, $\frac{1}{2}$ cup of sugar, and $\frac{1}{4}$ cup of chocolate chips. If there are only 12 cups of flour, 9 eggs, $4\frac{1}{4}$ cups of sugar, and $1\frac{3}{4}$ cups of chocolate chips available, how many total cookies can be made? (1 dozen = 12).

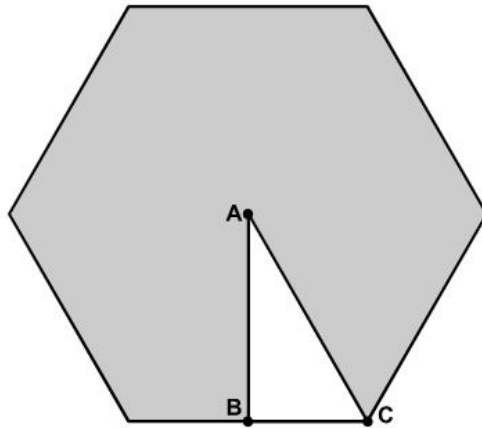
5. What is the sum of all numbers from 1 to 40 that are divisible by 4 or 5?

6. Ms. Mary baked more than 100 but less than 150 cupcakes for the school bake sale. If the number of cupcakes baked can be divided evenly into groups of 9, but will have one cupcake remaining when divided into groups of 5, how many cupcakes did Ms. Mary bake?

7. If a triangle has 0 diagonals, a square has 2 diagonals, a regular pentagon has 5 diagonals, and a regular hexagon has 9 diagonals, how many diagonals does a regular dodecagon (12 sides) have?



8. What is the 111th number in the arithmetic sequence: 3, 7, 11, 15...?
9. Five students, (Mary, James, Bob, Lewis, and Fred), are seated in chairs numbered 1 through 5. Mary may only sit in chair 1 or 5, James and Lewis may not sit next to each other, and Bob must be seated in any chair to the left of Fred. How many ways can the students be seated?
10. The distance from point A to point B is 4 in, and the distance from point B to point C is 3 in. What is the area of the shaded region, in square inches?



11. Tammy is taking a taxi service to visit her parents who live 64 miles away. Taxi service A charges \$25 for the first 10 miles and \$0.50 for each remaining mile. Taxi service B charges \$0.75 for each mile. What is the total cost of the cheaper taxi service, in dollars?
12. James wants to paint his room. His room has four walls measuring $15\text{ ft} \times 12\text{ ft}$, a ceiling measuring $15\text{ ft} \times 15\text{ ft}$, and a window measuring $4\text{ ft} \times 5\text{ ft}$. If 1 can of paint covers 90 ft^2 and costs \$15.00, what is the total cost of the total amount of paint cans James must purchase to paint his room, in dollars?
13. Jasmine is participating in a movie marathon starting at 11:45 AM. She watched four movies continuously with a duration of 105 minutes, 123 minutes, 98 minutes, and 132 minutes. If Jasmine only took two 15 minute breaks in between, what is the sum of the digits in Jasmine's end time?

14. The area of a circle is 72π and its circumference may be expressed as $a\sqrt{b\pi}$. What is the $a + b$?
15. A large $3 \times 3 \times 3$ cube, made up of 27 smaller unit cubes, is painted red on all 6 sides. The 27 smaller cubes are then mixed together and one cube is randomly selected. If the probability of the selected cube having red paint on two of its faces is a/b , what is the value of ab ?